ADVANCED NLP · MONSOON 2026

Which parts of BM25 help learned top-k selection?

Tests in sparse retrieval, KV-cache page selection and expert routing

Team ANLP Doomsday

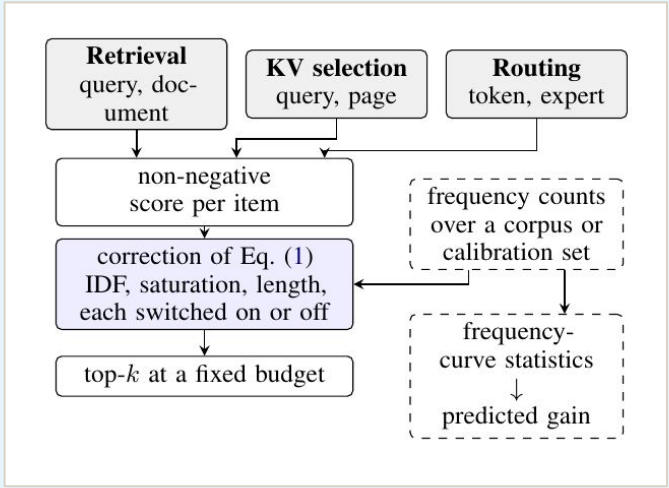


Figure 1 of our report: the three settings and where the BM25 correction sits

Three systems, one computation



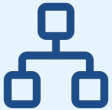
Retrieval

A query scores documents.
Keep the top-k documents.



KV-cache selection

A query scores KV pages (Quest).
Attend only to the top-k pages.



MoE routing

A token scores experts.
Send it to the top 8 of 64.

Only retrieval ever learned to correct the raw score. Does that correction travel, and can we tell when it helps?

BM25 is three small fixes to a raw score



IDF

Down-weight features that show up everywhere in the collection.



Saturation

Stop rewarding repeated evidence past a point.



Length

Neither favour nor punish long documents.

$$S(q, d) = \Sigma_j \, x_{qj} \cdot w_j(\mathrm{IDF}) \cdot T(x_{dj}, \ell_d)$$

Each fix is on or off, so 8 configurations. **000** = raw inner product, **111** = BM25.

Built for words, not for learned features.

Research disagree about whether it works

+0.20

nDCG@10 on sparse-autoencoder features of a dense retriever

−0.11

nDCG@10 on SPLADE-v3

The popular explanation: BM25 helps when feature frequencies are Zipf-like, as word frequencies are.

If that is true, the frequency curve should predict the gain before running anything. We tested exactly that.

Clavié et al. 2026 · Han et al. 2026

QUESTIONS AND ANSWERS

Four questions, four short answers

01	Does BM25 beat the raw inner product on learned sparse features?	**Yes: 162 of 179 rows**
02	Which of IDF, saturation and length explains the gain?	**Mostly saturation: 62%**
03	Can we predict the gain without relevance labels?	**No: frequency statistics fail**
04	Does it carry over to KV-cache selection and MoE routing?	**No: IDF breaks both**

STUDY DESIGN

One encoder, 14 sparse codes, 8 configurations



Text
queries and documents

Frozen E5
one dense encoder

Sparsify
7 bases, pooled or token

Score
8 BM25 configurations

nDCG@10
ranking quality

14
sparse representations

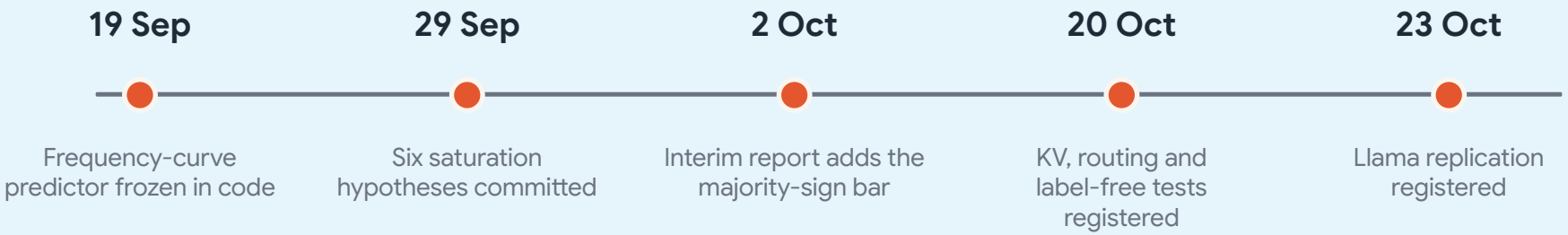
13
collections, English + Hindi, Bengali, Telugu

8
on/off configurations

1,358
rows, each with 8 nDCG@10 scores

Exact Shapley values split each gain. Paired bootstrap with Holm correction. Predictors validated on held-out families and corpora.

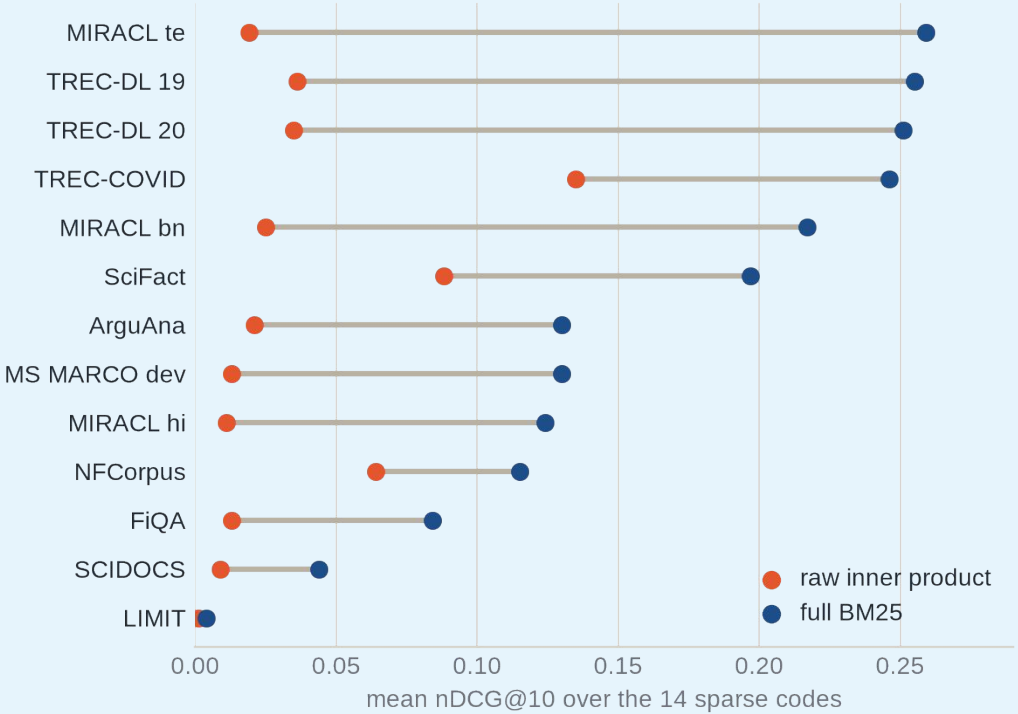
Pass marks were written down before any run



Six registered tests, each with a pass mark set in advance.

A prediction that can fail did fail, and we can say so.

BM25 beats the raw score in 162 of 179 rows



90%

of primary rows improve when BM25 replaces the raw inner product.

The best code (token-level SAE) beats text BM25 on 6 of 13 collections.

Caveat: the raw score is below 0.01 in 81 rows, so nearly any fix helps.

RQ2 · ATTRIBUTION

Most of the gain needs no collection statistics

Share of the raw → BM25 gain, exact Shapley values over 182 primary rows



Uses only the document's own values. No collection, no labels.

Second largest. Helps the E5 codes, but hurts SPLADE-v3.

The part the Zipf story is about contributes least, and shrinks as supports grow.

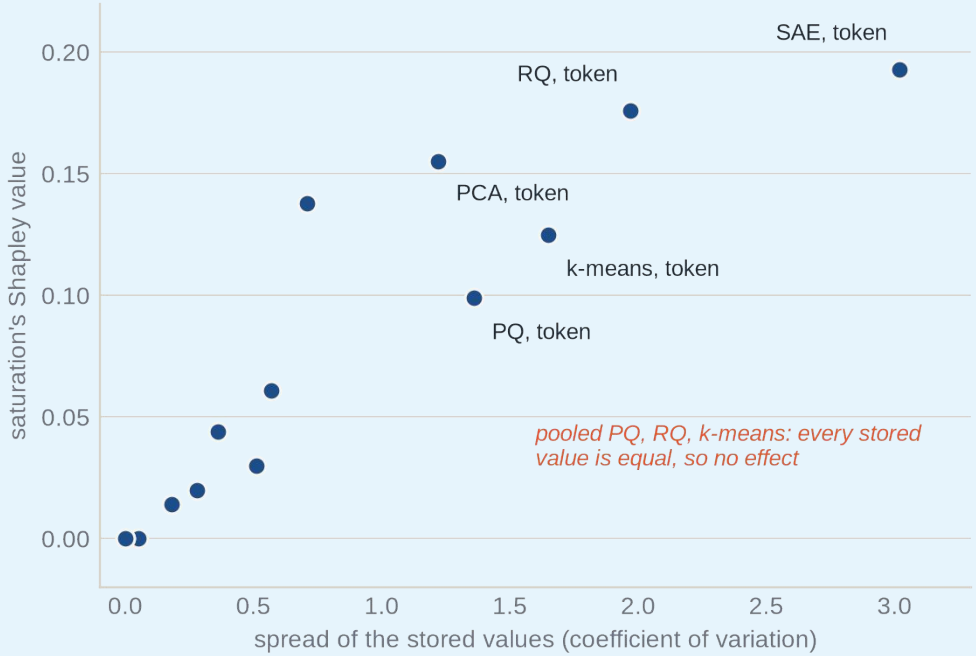
Collection statistics explain only 38% of the gain (95% CI 36 to 41%).

Saturation helps when stored values are uneven

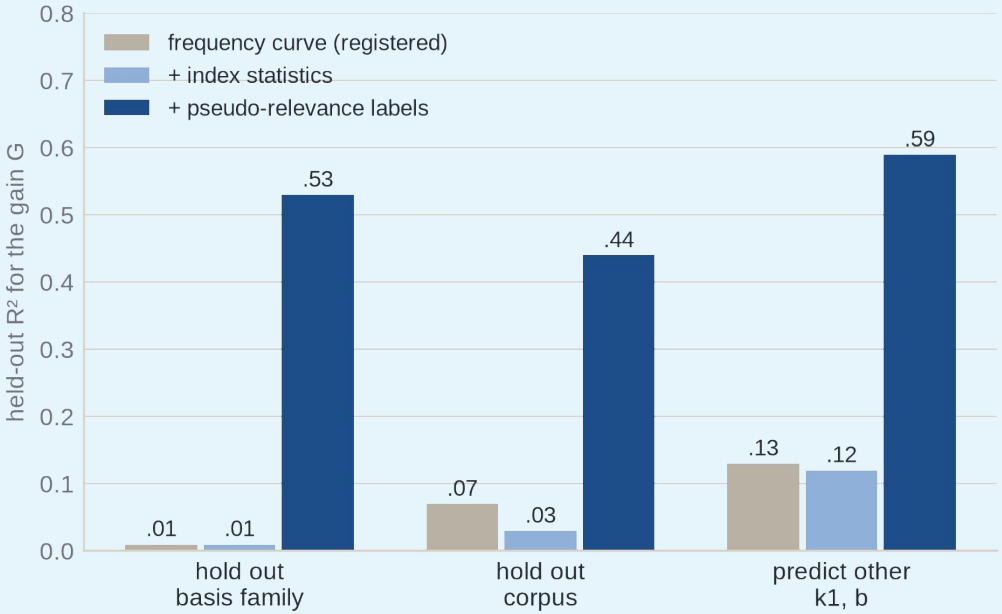
If every stored value were equal, saturation would rescale all scores by one factor and the ranking could not move.

More spread means more reordering. We registered this before testing it, and it held on settings we had not looked at.

Spearman 0.87 and 0.90, all six hypotheses pass.



Frequency statistics can't predict when BM25 helps

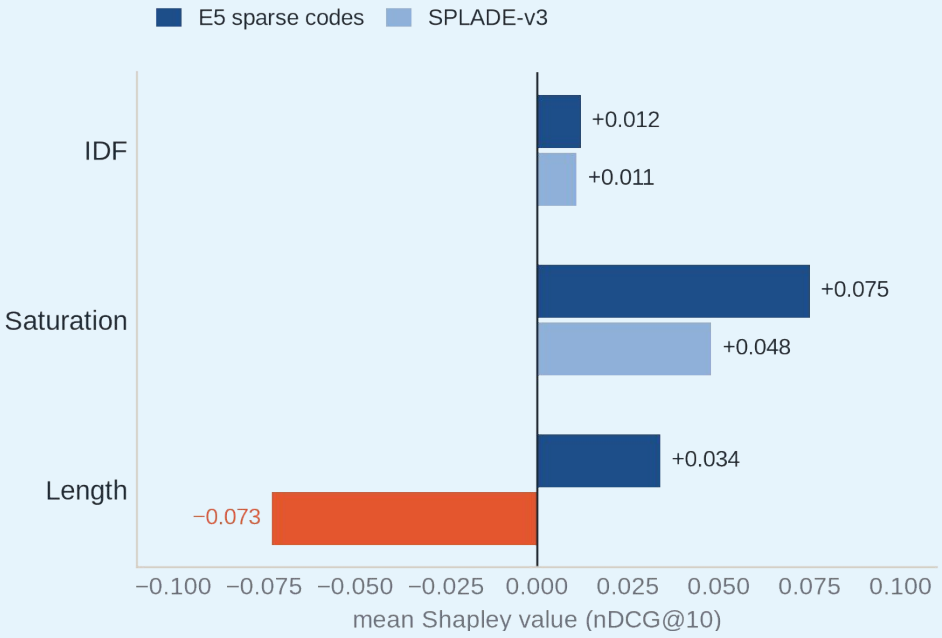


R^2 0.02

of the held-out variance explained by the Zipf-curve predictor. Its sign is right 90% of the time, exactly what "always say BM25 helps" scores.

Frequency says how common a feature is, not whether it matters.

On SPLADE-v3 the same correction hurts



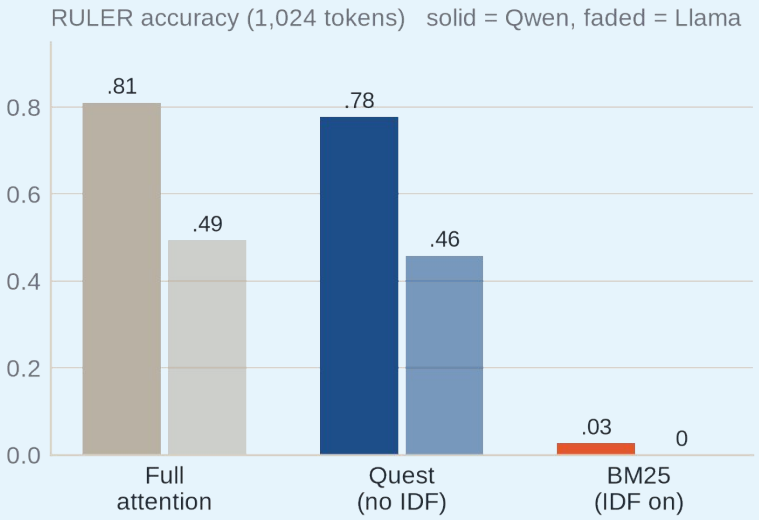
.523 → .509

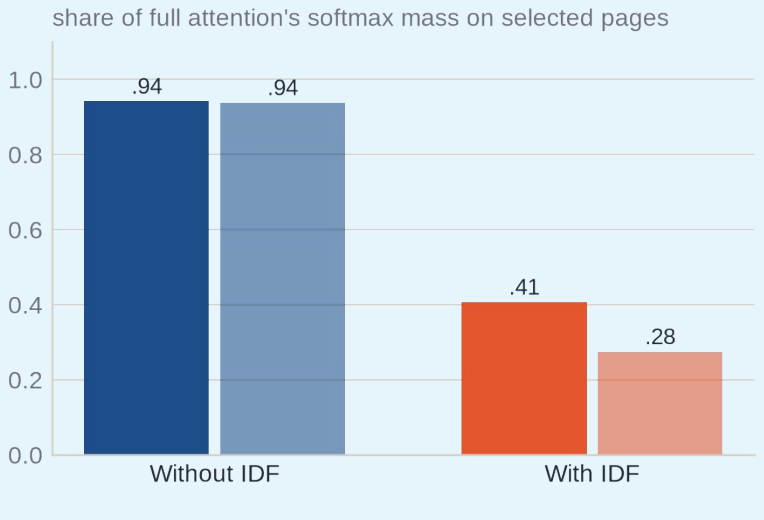
SPLADE-v3 nDCG@10 from the raw score to BM25. Worse on 8 of 10 collections, mostly through length normalisation.

Our label-free rule is right on only 5 of 10 collections here. Registered test: fail.

The answer depends on the representation, not just the method.

In KV-page selection, IDF collapses accuracy





IDF down-weights the pages Quest selects most often. Those are the pages the model needs. Worse on all 24 rows, better on none.

Honest caveat: partly by construction, since we define a page's frequency as its own selection rate.

MoE routing agrees, and why saturation can't help

PROPOSITION 1

With one score per item and IDF off, saturation selects the same top-k as no saturation.

Saturation is a strictly increasing function of f/ℓ, so it cannot reorder single-score items. Only IDF can, and here it reorders them wrongly.

✓ Confirmed: 001 to 011 match 000 on every example

OLMoE-1B-7B: MMLU accuracy over 57 subjects



WikiText-2 perplexity 7.49 → 7.88. Worse on 52 of 55 subjects.

SCORECARD

Six registered tests: 1 passed, 4 failed, 1 tied

✓	**Saturation mechanism (H1 to H6)**	k1 and b sweep	All six pass
✗	**Frequency-curve predictor**	retrieval, 179 rows	R^2 .04 and .02, sign no better than a constant
✗	**Label-free rule**	SPLADE-v3, 10 collections	sign .50 with the BM25 reference
✗	**Frequency-curve predictor**	KV pages, Qwen and Llama	sign .00, MAE 4.18 and 4.00 (limit 1.81)
⚖	**Label-free rule**	KV pages, Qwen and Llama	sign 1.00, but a constant also scores 1.00
✗	**Frequency-curve predictor**	routing, 57 subjects	sign .05, MAE 3.98

Negative results we can trust: the pass marks were fixed first.

TAKEAWAYS

What this means beyond our grid

1

Start with saturation

Cheap, label-free, and 62% of the gain in retrieval.

2

Don't port IDF blindly

What counts as "frequency" decides it. In KV and routing, IDF cost accuracy (.78 to .03, .48 to .38).

3

Zipf isn't enough

Frequency-curve statistics failed in retrieval, KV and routing. Relevance, not rarity, decides.

4

Test before you believe

Registered pass marks are why five disappointing results are still results.

Limits one encoder (E5), one 4K context with 100 examples per task, one MoE model. **Next** frequencies counted over unrelated prompts.

Thank you

Questions?

62%

of the gain is saturation

.78 → .03

Qwen accuracy once IDF is on

52 of 55

MMLU subjects worse with BM25

github.com/avi1o1/ANLP-Doomsday

Aviral Gupta · Mohit Kumar Singh · Anirudh Sankar · Arihant Tripathy